Pandas Practice Tasks: Duplicated and Sorting

Easy Level

1. Create a DataFrame with columns Name, Age, Marks where some rows are duplicated. Use df.duplicated() to identify duplicate rows.
2. From the same dataset, remove duplicate rows using df.drop_duplicates().
3. Create a DataFrame of students and sort it by Age using df.sort_values(by='Age').
4. Sort the same dataset in descending order of Marks.

Medium Level

1. Create a DataFrame with columns Name, City, Marks. Find duplicate rows based only on Name and City using df.duplicated(subset=['Name','City']).
2. Remove duplicate rows but keep the last occurrence using keep='last'.
3. Sort the DataFrame by Marks and then by Name.
4. Create a DataFrame with random index values like [3,1,4,2] and sort it using df.sort_index().

Hard Level

1. Create a dataset of students with Name, Subject, Marks where some rows repeat. Remove all duplicates using keep=False.
2. Create a dataset with Name, Age, Marks. Sort by Age (ascending) and if ages are equal then sort by Marks (descending).
3. Create a DataFrame with duplicate names and remove duplicates based only on the Name column.
4. Create a dataset with unordered index. Sort rows by index and then sort rows by Marks column. Explain the difference.

Bonus Real-World Task

1. Create a dataset with columns Name, Department, Salary.
2. Identify duplicate rows.
3. Remove duplicate rows.
4. Sort employees by Salary in descending order.
5. Sort employees by Department and Salary.